ASSIGNMENT–2

Foundations of Cryptography

Course: BCA | Subject: Cryptography

Q1. List the four fundamental cryptographic goals/objectives.

The four fundamental cryptographic goals are Confidentiality, Integrity, Authentication, and Non-repudiation. Together, these objectives help ensure that information remains private, accurate, originates from an authenticated party, and cannot easily be denied by its sender.

Q2. Define “Confidentiality” as a cryptographic goal with one example.

Confidentiality means protecting information from being accessed or disclosed to unauthorized people. For example, encryption used in online banking protects account and transaction information while it is transmitted over a network.

Q3. Explain Confidentiality, Integrity, Authentication, and Non-repudiation in detail with real-world scenarios. Why is it important to achieve all four together?

Confidentiality ensures that only authorized users can access the information. Encryption is a major mechanism for providing confidentiality. For example, encrypted banking communication prevents an attacker who intercepts network traffic from reading sensitive information.

Integrity ensures that information is not changed without authorization and that any unauthorized modification can be detected. Cryptographic hashes, message authentication mechanisms, and authenticated encryption can support integrity. For example, integrity protection can detect whether a file or message was modified during transmission.

Authentication verifies the identity of a user, device, or communicating system. Digital certificates and cryptographic authentication mechanisms are commonly used for this purpose. For example, during a secure web connection, a browser can use a server certificate to help authenticate the website.

Non-repudiation provides evidence that a particular party performed or approved an action, commonly through digital signatures. For example, a digitally signed electronic document can provide evidence that the holder of the corresponding private key signed it.

Achieving all four together is important because secure communication requires more than secrecy alone. Confidentiality protects the content, integrity detects unauthorized changes, authentication establishes who is communicating, and non-repudiation provides evidence of an action or signature. Weakness in any one of these areas can reduce the overall trustworthiness of a communication system.

Q4. What is the difference between a substitution cipher and a transposition cipher? Give one example of each.

A substitution cipher replaces each plaintext symbol with another symbol while keeping the general positions of the symbols unchanged. The Caesar Cipher is an example. A transposition cipher does not replace the symbols; instead, it rearranges their positions. The Rail Fence Cipher is an example of a transposition cipher.

Q5. Discuss the evolution from classical ciphers to modern cryptographic standards (DES → AES, RSA, SHA). Highlight key milestones and motivations for change.

Classical cryptography mainly used substitution and transposition techniques. Examples include the Caesar Cipher and Vigenère Cipher. These methods were historically useful but became vulnerable to systematic cryptanalysis and did not provide the security required by modern computer networks.

DES (Data Encryption Standard) represented an important transition to standardized computer-based symmetric encryption. It provided practical protection for digital data, but its 56-bit effective key size eventually became inadequate as computational power increased.

AES (Advanced Encryption Standard) was adopted to provide a stronger and more efficient modern symmetric encryption standard. AES supports larger key sizes and is widely used for protecting data in modern systems.

RSA introduced practical public-key cryptography using separate public and private keys. It helped address the key-distribution problem associated with symmetric cryptography and also enabled digital signatures.

SHA (Secure Hash Algorithm) represents a family of cryptographic hash functions used for integrity-related applications and other security purposes. Hashing differs from encryption because a secure cryptographic hash is designed as a one-way transformation rather than a reversible process.

The overall evolution was driven by increasing computational power, discovery of weaknesses in older algorithms, the need for secure key distribution, and the growing volume and importance of digital information.

Q6. What is a “key” in cryptography, and why is key management important?

A cryptographic key is a value used by a cryptographic algorithm to control encryption, decryption, authentication, or signing operations. Key management is important because keys must be generated, stored, distributed, rotated, protected, and eventually revoked or destroyed securely; a compromised key can undermine the protection provided by an otherwise strong algorithm.

Q7. Explain how cryptographic techniques protect data during transmission, prevent unauthorized access, and detect tampering in a network.

Cryptography protects network communication by applying encryption, authentication, and integrity mechanisms.

During transmission, encryption converts readable data into ciphertext so that an interceptor cannot easily understand the information. For example, TLS protects data exchanged between a web browser and a secure website.

Cryptographic authentication helps prevent unauthorized access by verifying users, servers, devices, or services. Digital certificates and cryptographic authentication protocols can establish trust in the identity of a communicating endpoint.

Integrity mechanisms help detect tampering. A cryptographic hash, MAC, or authenticated-encryption mechanism can allow the receiver to detect whether protected data has been altered. For example, if the integrity verification fails, a received message can be rejected rather than trusted.

Therefore, modern network security generally combines confidentiality, authentication, and integrity rather than depending on encryption alone.

Q8. Differentiate between a passive attack and an active attack on a cryptographic system, with one example of each.

A passive attack observes or collects information without directly changing the communication. For example, an attacker may eavesdrop on network traffic to observe unprotected information. An active attack involves modifying, injecting, deleting, or disrupting information. A man-in-the-middle attacker who alters messages between two communicating parties is an example of an active attack.

Q9. Describe how cryptographic techniques evolved historically to counter emerging security threats. Include the shift from symmetric-only systems to asymmetric cryptography and how this addressed the key-distribution problem.

Early cryptographic systems mainly relied on symmetric techniques, in which the same secret key was used for encryption and decryption. Although efficient, symmetric cryptography created a major key-distribution problem: communicating parties needed a secure method to share the secret key before secure communication could begin.

As communication networks expanded, securely distributing a different secret key to many users became increasingly difficult. The development of public-key cryptography provided a major change. Instead of one shared secret key, asymmetric cryptography uses a mathematically related public key and private key.

A public key can be distributed openly, while the corresponding private key remains secret. This makes it possible to establish secure communication or perform authentication without first sending a shared secret key through the same communication channel. Public-key techniques also support digital signatures.

Modern security systems commonly combine both approaches: asymmetric cryptography is used for functions such as authentication and key establishment, while faster symmetric cryptography is used to protect the actual bulk data.

Q10. Give one example each of a modern symmetric algorithm and a modern asymmetric algorithm used today.

A modern symmetric algorithm is AES (Advanced Encryption Standard). A modern asymmetric algorithm is RSA (Rivest–Shamir–Adleman).

Q11. Discuss the classification of cryptography into Classical Cryptography and Modern Cryptography. Explain the transition from manual/mechanical methods to computer-based algorithms, mentioning key milestones and technologies involved.

Classical Cryptography refers to early techniques that were generally performed manually or with relatively simple mechanical systems. Substitution methods such as the Caesar Cipher and Vigenère Cipher changed symbols, while transposition methods rearranged their positions. These techniques were useful for historical communication but had limitations against systematic cryptanalysis.

Modern Cryptography is based mainly on mathematical algorithms implemented by computers. It includes symmetric encryption, asymmetric or public-key cryptography, cryptographic hashing, digital signatures, and secure communication protocols.

A major milestone was the development of complex mechanical and electromechanical systems during World War II, including the Enigma machine. The growth of computers then made it possible to design and analyze much more sophisticated cryptographic algorithms.

DES became an important standardized computer-based symmetric encryption algorithm. The later development and adoption of AES provided a stronger modern symmetric standard. The introduction of public-key cryptography, including RSA, addressed important key-distribution and authentication requirements. SHA-family hash functions added widely used cryptographic hashing capabilities.

Thus, the transition from classical to modern cryptography was driven by computers, mathematical cryptanalysis, increasing communication needs, greater computational power, and the requirement for stronger and scalable security systems.